Q Pathformer routing capacity and classical deployment

Version 07 internal study findings

Prepared for Anthony Siljeg | QPATH RPT008 | 21 October 2026 | Internal

Findings and decision

Geometric routing remains a useful classical mechanism to develop. Across fresh replicated populations, grouping nearby inputs into specialist banks consistently outperformed random hash routing with exactly matched training bank counts. This is the clearest Q-Pathformer component finding from v0.7.

The two-qubit circuit is a working research component, but the strongest classical control still led on the two harder clean tasks. The global 121-feature trigonometric model averaged 98.47% macro-F1 on checkerboard and 96.57% on spiral. The flexible routed circuit averaged 76.57% and 86.82%. Routing geometry and the choice of feature basis both matter; these results do not validate the full proposed architecture.

Task	Flexible circuit	Routed Trig25	Global Trig121
Checkerboard	76.57 ± 2.43	90.18 ± 1.17	98.47 ± 0.20
Spiral	86.82 ± 3.68	92.42 ± 1.23	96.57 ± 0.45
Noisy annulus	87.14 ± 1.13	87.75 ± 0.94	87.28 ± 1.25
Rare disks	98.70 ± 0.31	98.94 ± 0.30	99.16 ± 0.51

Macro-F1 percentages, mean ± sample SD of eight population means; two training seeds are averaged within each population. These are descriptive results, not the primary BCE comparisons or a test-based model selection rule.

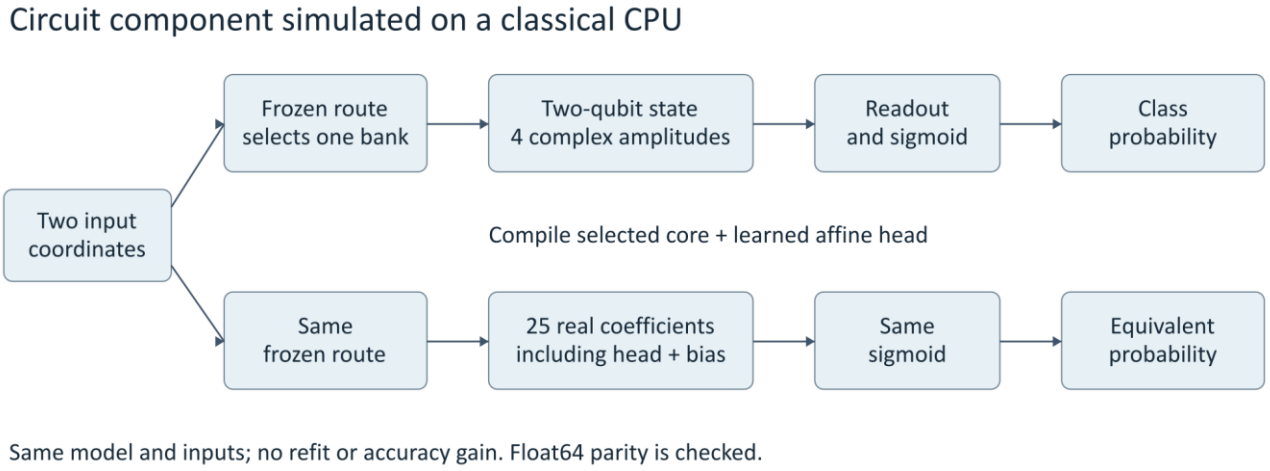
Classical compilation preserved every checked class prediction and made single-input circuit calls 4.28–4.58 times faster across task and arm population medians. It preserves the selected model's accuracy. The fastest conventional control remained the MLP, with weaker predictive scores; the richer global trigonometric control offered a stronger accuracy and cost comparison. Flexible circuit heads remained sensitive to finite-shot sampling, so classical accuracy alone is not a hardware-readiness result.

All 608 registered fits finished without a training failure or retry. The 512 confirmation endpoints were audited before test release, and independent numerical replay confirmed all predictions and the 20 registered primary comparisons. The following sections separate predictive quality, deployment costs and local measurement sensitivity.

The objective is a useful classical implementation and evidence for future quantum testing; this study does not assess quantum advantage. Retain geometric and random routing controls and the richer global classical control, then test one distinctive architecture mechanism. Physical quantum comparisons remain planned for paper v1.1.

What was tested

The circuit arms explicitly emulate a two-qubit state on a classical CPU: four complex amplitudes are transformed by gates, and measured expectations feed a learned classifier. One frozen input router selects one of five parameter banks for each example. There is no physical quantum device in this study.



The compiled path includes the selected circuit core and learned head and bias. It changes how the selected function is evaluated, not what was trained. Independently trained trigonometric and neural controls provide the comparison.

Arm	Params	Route	Role
ScalarK	50	5 kmeans banks	One circuit expectation and affine head
MultiK	70	5 kmeans banks	Five circuit expectations and affine head
MultiHash	70	5 hash banks	Matched-count routing control for MultiK
TrigK	125	5 kmeans banks	25 learned trigonometric coefficients per bank
TrigHash	125	5 hash banks	Matched-count routing control for TrigK
TrigG25	25	Global	Frequency 2 on each input axis
TrigG121	121	Global	Frequency 5 on each input axis
MLPG129	129	Global	Conventional 32-unit ReLU hidden layer

Each circuit bank has two layers, input RY encoding, trainable RY and RZ rotations and one CX gate per layer, with eight trainable core angles. ScalarK reads X0. The flexible head reads X0, X1, Z0, Z1 and Z0Z1. Parameter counts exclude routing metadata; similar counts do not imply identical function classes or optimization difficulty.

Fresh data and comparison design

The four binary classification tasks use two coordinates sampled uniformly from $[-1,1]^2$: a periodic checkerboard, a winding spiral, a noisy annulus and three rare positive disks. Annulus labels are independently flipped with probability 0.10. The primary results use observed labels; clean labels never enter training or selection.

Stage	Populations	Seeds per population	Fits	Updates per fit
Engineering	4	1	32	150
Development A	4	2	64	15,000
Confirmation B	32	2	512	15,000

Each of 40 populations has 8,000 records: 4,800 training, 1,600 validation and 1,600 test. All 320,000 new coordinates were checked for exact uniqueness and excluded against 294,400 retained historical coordinates including fixtures. This establishes coordinate freshness, not semantic novelty or performance on unrelated data.

Confirmation contains eight independently generated populations per task, each with paired training seeds 1401 and 1402. Kmeans routers are fitted anew per population on training coordinates only. The hash router's four training-rank cutpoints exactly match the corresponding five kmeans training counts. The same cutpoints are used on validation and test without rebalancing. Equal training occupancy does not make held-out banks or their class composition equivalent.

Every full fit completed 200 rounds and 15,000 Adam updates, with batch size 64, learning rate 0.01, betas 0.9/0.999 and epsilon 1e-8. All arms optimized native-logit binary cross-entropy. There was no early stopping, scheduler, gradient clipping, weight decay or outcome-based arm removal. The 576 full fits consumed 8.64 million updates and 552.96 million training-record visits; the engineering fits add 4,800 updates.

The selected checkpoint is the earliest exact minimum validation BCE over the full horizon, using natural logarithms and probabilities clipped to [1e-7,1-1e-7]. Test labels never select an epoch, route, threshold or model. A probability of at least 0.5 predicts the positive class.

For inference, the two seed differences are averaged within each population, leaving eight independent values per task. The 20 primary BCE comparisons use nominal 99.75% Bonferroni-adjusted t intervals with seven degrees of freedom. Negative differences favor the first named arm. A zero-crossing interval is inconclusive, not equivalence. The predeclared practical reference is 0.01 BCE; small-sample coverage depends on distributional assumptions.

Hardware and execution

Item	Recorded configuration
Computer	MSI Stealth A18 AI+ A3XWJG
Processor	AMD Ryzen AI 9 HX 370; 12 physical cores, 24 logical processors
Memory	67,772,469,248 Windows-reported physical-memory bytes, approximately 63.12 GiB
Operating system	64-bit Windows 11 Pro, build 26300
Scientific runtime	Python 3.12.14; NumPy 2.5.4; SciPy 1.18.1
Numerical library	OpenBLAS 0.3.34.106.0; one effective BLAS thread per worker
Devices used	CPU only; no GPU training or physical quantum jobs
Execution layer	Q-Path Multiplier (ETEL), based on ETEL 0.2.0; at most five concurrent fits

ETEL schedules complete registered fits in isolated processes. Persistent claims, process creation identities, immutable attempt directories and committed hashes support restart reconciliation and prevent duplicate completed jobs. Its logical groups of 64 are bookkeeping; they do not represent 64 active processors, model banks or a measured 64-fold speedup.

All 608 worker environment receipts confirm one effective BLAS thread and no GPU use. Each stage reached five owned workers. The largest individual worker peak working set was 61.76 MiB; this is not a measurement of simultaneous total machine memory. Training used cached inputs and features where the frozen implementation permits them; deployment timing separately measures complete raw-input calls.

Full-horizon training, shared preparation charges, process CPU, overlapping fit wall timers, stage supervision elapsed time and isolated inference latency have different meanings. The resource section keeps them separate. The device inventory does not measure energy use, FLOPs, thermal stability or quantum performance.

How this extends the sealed baseline

The v0.1 to v0.6 development sequence motivated this component study. The immediate sealed v0.6 comparison used the same four task formulas, six arms and one confirmation population per task with five optimization seeds. V0.7 adds independent population replication, matched-count random routing controls, the wider global Trig121 control, three interleaved timing sessions and equal total simulated measurement budgets.

v0.6 arm	Checkerboard	Spiral	Noisy annulus	Rare disks
ScalarK	73.03 / .5780	77.60 / .5007	86.20 / .4106	99.12 / .0135
MultiK	78.55 / .5216	84.71 / .4253	86.92 / .4091	98.85 / .0146
TanhK	60.02 / .6511	66.05 / .5682	83.07 / .4322	94.17 / .0523
TrigK	89.57 / .3050	92.05 / .2483	86.81 / .4032	98.67 / .0149
TrigG25	48.89 / .6965	57.57 / .6679	82.14 / .4597	99.30 / .0167
MLPG129	69.55 / .5667	75.77 / .4695	85.28 / .4184	95.12 / .0523

Historical v0.6 means: macro-F1 percentage / clipped BCE, across five seeds on one population per task. All six controls are shown. These values are not pooled with v0.7 and raw cross-version changes are not estimates of improvement.

V0.6 suggested that a flexible readout could improve the circuit while directly learned routed trigonometric coefficients remained stronger on checkerboard and spiral. The high rare-disk scores also showed why reporting a best F1 alone is insufficient. V0.7 addresses these questions with controlled routing and broader population-level evidence.

Historical v0.6 conversion preserved 40 selected circuit functions, with maximum probability difference 7.16e-15 and no class disagreement. Its timing came from one session with observed block drift. Its 1,024/8,192 measurement budgets were per setting, so flexible heads used twice the total of scalar heads. V0.7 timings and shot errors must therefore be interpreted under their own revised protocols.

The complete historical context capsule retains all values and source bindings under report_work/final_report/history. Earlier versions remain preserved; this report does not revise the architecture proposal or the internal paper draft.

Predictive results at a glance

Held-out macro-F1 across fresh populations

	Checkerboard	Spiral	Noisy annulus	Rare disks
ScalarK	72.08	80.86	86.42	98.70
MultiK	76.57	86.82	87.14	98.70
MultiHash	49.95	56.80	78.41	91.89
TrigK	90.18	92.42	87.75	98.94
TrigHash	49.40	57.72	81.91	97.74
TrigG25	50.89	58.74	83.17	99.00
TrigG121	98.47	96.57	87.28	99.16
MLPG129	65.98	71.68	86.38	91.67

Means over 8 populations per task; 2 paired seeds averaged within each.
F1 is descriptive. Tables give population SDs. Blue scale spans 0 to 100%.

High scores depend on the task and representation. The richer global trigonometric control leads on checkerboard and spiral, while several models approach the ceiling on rare disks. The noisy annulus remains harder to classify perfectly because its observed labels contain independent flips.

The following tables include every registered arm. Their population standard deviations describe variation. Statistical direction is assessed by the separately registered paired BCE comparisons, rather than this figure.

Checkerboard predictive performance

Entries are means with population sample SD in parentheses. Each population first averages its two paired training seeds; the eight fresh populations supply the independent replication. All eight arms remain in the report. Lower BCE, Brier and ECE are better.

Arm	Common BCE	Macro F1 %	Accuracy %	Brier	ECE10
ScalarK	0.552 (0.014)	72.08 (1.57)	72.12 (1.59)	0.184 (0.005)	0.062 (0.018)
MultiK	0.509 (0.025)	76.57 (2.43)	76.59 (2.42)	0.167 (0.009)	0.071 (0.018)
MultiHash	0.699 (0.002)	49.95 (1.22)	50.06 (1.20)	0.253 (0.001)	0.041 (0.010)
TrigK	0.266 (0.015)	90.18 (1.17)	90.18 (1.17)	0.078 (0.005)	0.071 (0.011)
TrigHash	0.698 (0.002)	49.40 (1.33)	49.50 (1.33)	0.253 (0.001)	0.045 (0.013)
TrigG25	0.696 (0.002)	50.89 (2.28)	51.24 (2.12)	0.251 (0.001)	0.024 (0.015)
TrigG121	0.050 (0.005)	98.47 (0.20)	98.47 (0.20)	0.014 (0.001)	0.021 (0.003)
MLPG129	0.599 (0.020)	65.98 (2.51)	66.23 (2.29)	0.207 (0.008)	0.063 (0.010)

Common clipped BCE is the registered primary outcome. Macro F1, accuracy and calibration summaries are descriptive; the highest F1 alone does not select a model. Macro F1 averages both fixed classes; a class with a zero F1 denominator contributes zero.

Positive class and native loss

Arm	Native BCE	Precision %	Recall %	Grouped AP %
ScalarK	0.552 (0.014)	72.34 (1.30)	71.42 (4.75)	79.57 (1.75)
MultiK	0.509 (0.025)	76.20 (2.91)	77.29 (3.27)	83.53 (2.91)
MultiHash	0.699 (0.002)	49.87 (1.27)	47.98 (3.94)	50.24 (1.32)
TrigK	0.266 (0.015)	89.67 (1.28)	90.76 (1.37)	96.59 (0.71)
TrigHash	0.698 (0.002)	49.31 (1.53)	50.11 (4.52)	49.73 (1.48)
TrigG25	0.696 (0.002)	51.24 (2.16)	46.41 (5.95)	49.61 (1.63)
TrigG121	0.050 (0.005)	98.41 (0.48)	98.52 (0.38)	99.92 (0.02)
MLPG129	0.599 (0.020)	67.42 (2.25)	63.22 (8.54)	73.33 (2.42)

Positive precision and recall use the fixed 0.5 threshold. Grouped AP treats equal scores as tied groups. Undefined ratios remain undefined in the complete evidence. Native BCE uses logits; the registered comparisons use common clipped BCE.

The high frequency global trigonometric control has the strongest descriptive scores here. Geometric routing helps the low bandwidth models, but does not close the basis gap.

Complete population values and undefined identities: analysis/official_results/stage_B_v01/descriptive_summary.json

Spiral predictive performance

Entries are means with population sample SD in parentheses. Each population first averages its two paired training seeds; the eight fresh populations supply the independent replication. All eight arms remain in the report. Lower BCE, Brier and ECE are better.

Arm	Common BCE	Macro F1 %	Accuracy %	Brier	ECE10
ScalarK	0.450 (0.046)	80.86 (2.93)	80.87 (2.92)	0.137 (0.017)	0.056 (0.009)
MultiK	0.372 (0.067)	86.82 (3.68)	86.82 (3.67)	0.103 (0.026)	0.065 (0.011)
MultiHash	0.671 (0.006)	56.80 (1.47)	56.85 (1.43)	0.240 (0.003)	0.044 (0.008)
TrigK	0.250 (0.044)	92.42 (1.23)	92.42 (1.23)	0.062 (0.010)	0.046 (0.017)
TrigHash	0.664 (0.005)	57.72 (1.30)	57.74 (1.30)	0.237 (0.002)	0.045 (0.017)
TrigG25	0.655 (0.005)	58.74 (0.91)	58.77 (0.92)	0.233 (0.002)	0.052 (0.013)
TrigG121	0.079 (0.005)	96.57 (0.45)	96.57 (0.45)	0.024 (0.002)	0.012 (0.003)
MLPG129	0.517 (0.040)	71.68 (4.12)	71.73 (4.13)	0.172 (0.018)	0.062 (0.010)

Common clipped BCE is the registered primary outcome. Macro F1, accuracy and calibration summaries are descriptive; the highest F1 alone does not select a model. Macro F1 averages both fixed classes; a class with a zero F1 denominator contributes zero.

Positive class and native loss

Arm	Native BCE	Precision %	Recall %	Grouped AP %
ScalarK	0.450 (0.045)	81.51 (3.24)	79.91 (3.09)	88.43 (2.39)
MultiK	0.373 (0.065)	87.14 (3.28)	86.36 (4.63)	92.53 (3.21)
MultiHash	0.671 (0.006)	57.06 (1.63)	55.24 (3.26)	61.96 (1.61)
TrigK	0.255 (0.044)	92.40 (1.18)	92.44 (1.62)	96.87 (0.80)
TrigHash	0.664 (0.005)	57.78 (1.46)	57.65 (2.06)	62.98 (0.77)
TrigG25	0.655 (0.005)	58.76 (1.00)	58.86 (2.81)	64.69 (1.04)
TrigG121	0.079 (0.005)	96.64 (0.71)	96.50 (0.44)	99.64 (0.05)
MLPG129	0.517 (0.040)	72.27 (4.77)	70.91 (4.90)	82.29 (3.76)

Positive precision and recall use the fixed 0.5 threshold. Grouped AP treats equal scores as tied groups. Undefined ratios remain undefined in the complete evidence. Native BCE uses logits; the registered comparisons use common clipped BCE.

The global high frequency control also leads this task. The routed trigonometric model improves on the circuit readout, showing that readout flexibility does not remove every capacity limit.

Complete population values and undefined identities: analysis/official_results/stage_B_v01/descriptive_summary.json

Noisy annulus predictive performance

Entries are means with population sample SD in parentheses. Each population first averages its two paired training seeds; the eight fresh populations supply the independent replication. All eight arms remain in the report. Lower BCE, Brier and ECE are better.

Arm	Common BCE	Macro F1 %	Accuracy %	Brier	ECE10
ScalarK	0.406 (0.020)	86.42 (1.11)	86.46 (1.10)	0.119 (0.007)	0.068 (0.010)
MultiK	0.399 (0.020)	87.14 (1.13)	87.19 (1.13)	0.116 (0.007)	0.074 (0.010)
MultiHash	0.514 (0.007)	78.41 (0.36)	78.42 (0.36)	0.163 (0.002)	0.066 (0.008)
TrigK	0.394 (0.021)	87.75 (0.94)	87.80 (0.94)	0.114 (0.007)	0.076 (0.011)
TrigHash	0.466 (0.016)	81.91 (0.82)	81.92 (0.82)	0.142 (0.006)	0.063 (0.006)
TrigG25	0.454 (0.017)	83.17 (0.84)	83.18 (0.84)	0.136 (0.006)	0.069 (0.006)
TrigG121	0.395 (0.020)	87.28 (1.25)	87.34 (1.23)	0.114 (0.007)	0.073 (0.010)
MLPG129	0.427 (0.021)	86.38 (1.17)	86.41 (1.18)	0.123 (0.006)	0.091 (0.009)

Common clipped BCE is the registered primary outcome. Macro F1, accuracy and calibration summaries are descriptive; the highest F1 alone does not select a model. Macro F1 averages both fixed classes; a class with a zero F1 denominator contributes zero.

Positive class and native loss

Arm	Native BCE	Precision %	Recall %	Grouped AP %
ScalarK	0.406 (0.020)	86.05 (1.33)	85.19 (1.58)	85.50 (1.63)
MultiK	0.399 (0.020)	86.86 (1.25)	85.91 (1.40)	85.89 (1.70)
MultiHash	0.514 (0.007)	74.34 (0.75)	83.06 (1.56)	79.11 (1.51)
TrigK	0.394 (0.021)	88.06 (1.13)	85.84 (1.43)	86.21 (1.68)
TrigHash	0.466 (0.016)	78.56 (0.87)	84.96 (1.28)	83.40 (2.02)
TrigG25	0.454 (0.017)	79.85 (1.04)	86.17 (1.09)	84.65 (2.06)
TrigG121	0.395 (0.020)	87.79 (1.29)	85.05 (1.91)	86.23 (1.60)
MLPG129	0.427 (0.021)	84.56 (1.74)	87.23 (1.67)	85.76 (1.66)

Positive precision and recall use the fixed 0.5 threshold. Grouped AP treats equal scores as tied groups. Undefined ratios remain undefined in the complete evidence. Native BCE uses logits; the registered comparisons use common clipped BCE.

This task includes label noise. Scores near 97 percent are not an expected target; small differences among the stronger arms need the registered paired intervals on the following comparison pages.

Complete population values and undefined identities: analysis/official_results/stage_B_v01/descriptive_summary.json

Rare disks predictive performance

Entries are means with population sample SD in parentheses. Each population first averages its two paired training seeds; the eight fresh populations supply the independent replication. All eight arms remain in the report. Lower BCE, Brier and ECE are better.

Arm	Common BCE	Macro F1 %	Accuracy %	Brier	ECE10
ScalarK	0.018 (0.005)	98.70 (0.56)	99.47 (0.23)	0.005 (0.001)	0.009 (0.003)
MultiK	0.016 (0.003)	98.70 (0.31)	99.47 (0.13)	0.004 (0.001)	0.008 (0.001)
MultiHash	0.085 (0.007)	91.89 (0.86)	96.79 (0.37)	0.024 (0.002)	0.013 (0.003)
TrigK	0.013 (0.002)	98.94 (0.30)	99.57 (0.13)	0.004 (0.001)	0.006 (0.001)
TrigHash	0.025 (0.003)	97.74 (0.48)	99.07 (0.20)	0.007 (0.001)	0.008 (0.002)
TrigG25	0.016 (0.002)	99.00 (0.42)	99.59 (0.17)	0.004 (0.001)	0.009 (0.002)
TrigG121	0.008 (0.002)	99.16 (0.51)	99.66 (0.21)	0.002 (0.001)	0.004 (0.000)
MLPG129	0.074 (0.027)	91.67 (4.11)	96.88 (1.37)	0.023 (0.009)	0.021 (0.011)

Common clipped BCE is the registered primary outcome. Macro F1, accuracy and calibration summaries are descriptive; the highest F1 alone does not select a model. Macro F1 averages both fixed classes; a class with a zero F1 denominator contributes zero.

Positive class and native loss

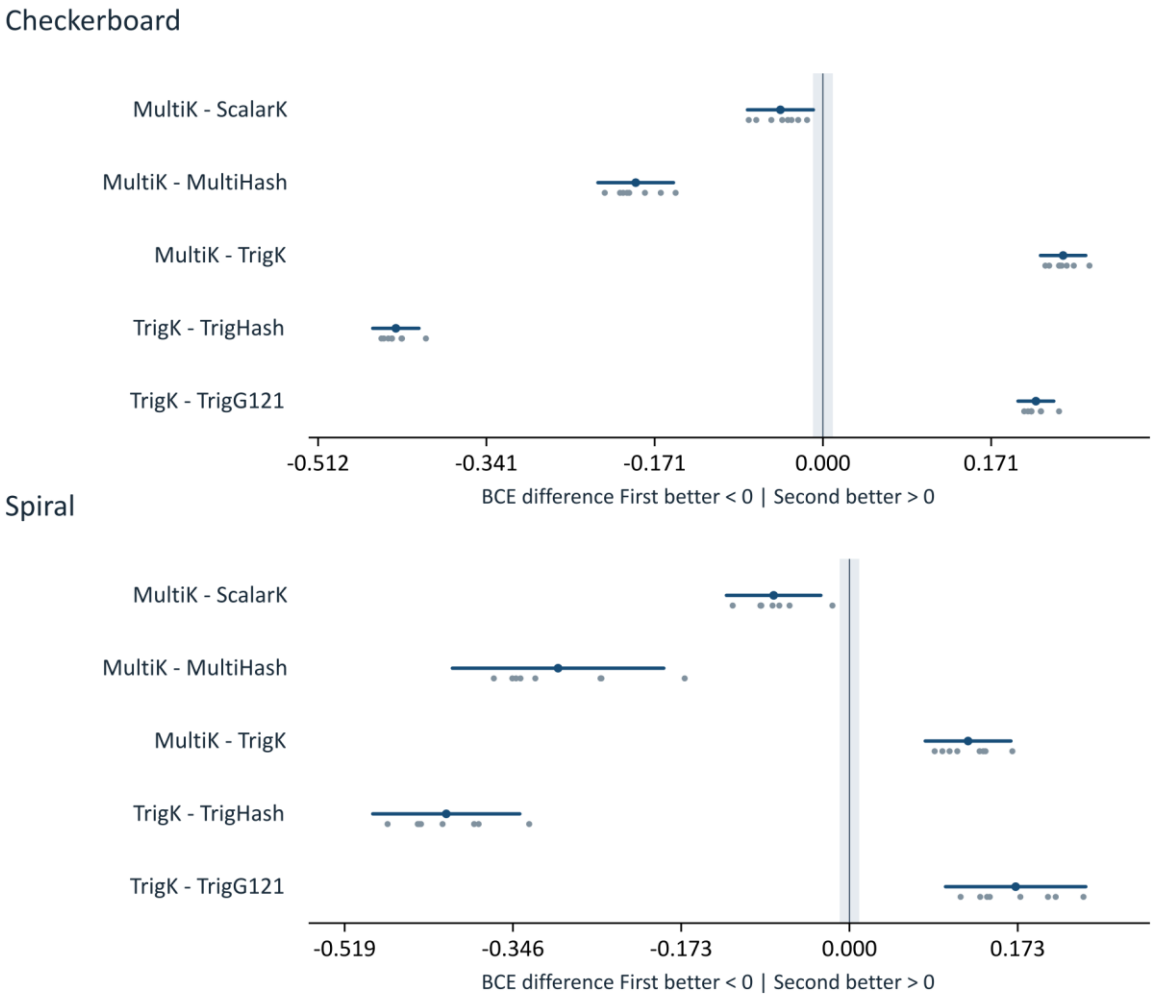
Arm	Native BCE	Precision %	Recall %	Grouped AP %
ScalarK	0.018 (0.005)	98.23 (0.90)	97.20 (1.48)	99.82 (0.13)
MultiK	0.016 (0.003)	97.85 (1.07)	97.57 (1.43)	99.86 (0.06)
MultiHash	0.085 (0.007)	88.91 (2.39)	82.57 (2.32)	93.85 (1.09)
TrigK	0.013 (0.002)	98.51 (0.88)	97.73 (0.95)	99.89 (0.07)
TrigHash	0.025 (0.003)	95.87 (1.70)	96.17 (1.71)	99.49 (0.19)
TrigG25	0.016 (0.002)	98.25 (1.18)	98.21 (1.00)	99.91 (0.06)
TrigG121	0.008 (0.002)	98.67 (0.68)	98.38 (1.44)	99.94 (0.05)
MLPG129	0.074 (0.027)	90.52 (5.48)	81.43 (10.06)	93.59 (4.88)

Positive precision and recall use the fixed 0.5 threshold. Grouped AP treats equal scores as tied groups. Undefined ratios remain undefined in the complete evidence. Native BCE uses logits; the registered comparisons use common clipped BCE.

This task is imbalanced. Accuracy alone can conceal missed positives, so macro F1, positive recall and AP are shown together. Near ceiling F1 does not imply equal BCE.

Complete population values and undefined identities: analysis/official_results/stage_B_v01/descriptive_summary.json

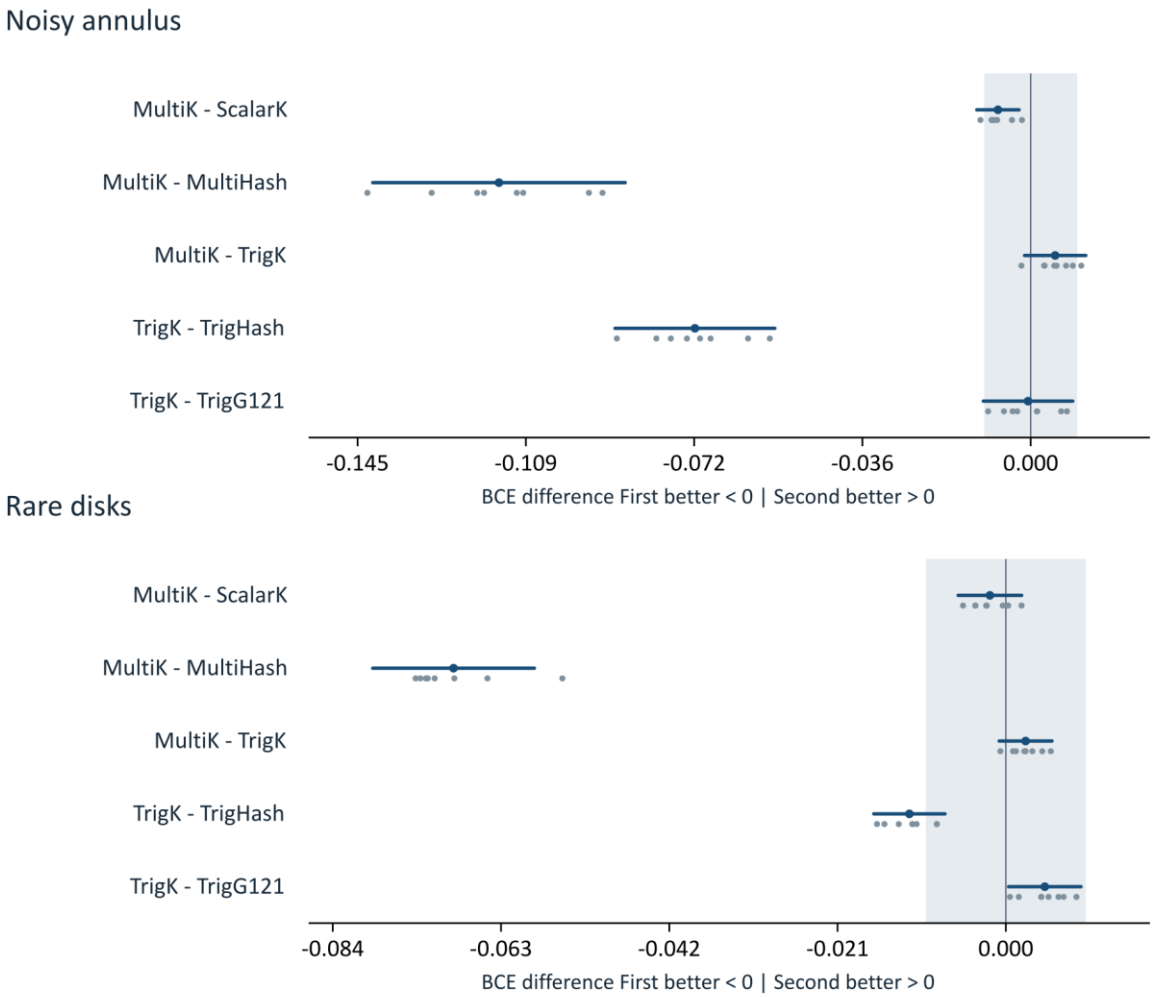
Primary comparison intervals on the clean tasks



Bars: nominal 99.75% t intervals, df 7. Grey dots: 8 population means.
Two paired seed differences are averaged within each population.
Shading: +/-0.01 practical reference, not equivalence.
Small-sample assumptions apply; repetitions do not increase n.

Each grey point averages two paired seed differences within one population. The shaded 0.01 BCE band is a practical reference, not an equivalence test. Complete numeric values follow.

Primary comparison intervals under noise and imbalance



Bars: nominal 99.75% t intervals, df 7. Grey dots: 8 population means.
Two paired seed differences are averaged within each population.
Shading: +/-0.01 practical reference, not equivalence.
Small-sample assumptions apply; repetitions do not increase n.

Each grey point averages two paired seed differences within one population. The shaded 0.01 BCE band is a practical reference, not an equivalence test. Complete numeric values follow.

Checkerboard primary comparisons

Differences are first arm minus second arm common clipped BCE; negative values favor the first. Each interval uses eight population differences after averaging the two paired seeds. The nominal 99.75 percent intervals apply the registered Bonferroni family of 20 comparisons with 7 degrees of freedom.

Comparison	Mean	Interval	Interpretation
C1 MultiK versus ScalarK	-0.04309	[-0.07648, -0.00969]	First lower 0.01 not established
C2 MultiK versus MultiHash	-0.18977	[-0.22805, -0.15148]	First lower Beyond 0.01
C3 MultiK versus TrigK	0.24348	[0.22054, 0.26643]	Second lower Beyond 0.01
C4 TrigK versus TrigHash	-0.43289	[-0.45630, -0.40949]	First lower Beyond 0.01
C5 TrigK versus TrigG121	0.21588	[0.19777, 0.23399]	Second lower Beyond 0.01

Beyond 0.01 means the entire interval exceeds the predeclared practical reference in the stated direction. Crossing zero is inconclusive, not evidence of equivalence. Small sample interval assumptions remain unverified.

All population differences

Population	C1	C2	C3	C4	C5
1	-0.02506	-0.18039	0.24718	-0.42668	0.22083
2	-0.05227	-0.19638	0.24207	-0.43688	0.21119
3	-0.07519	-0.22114	0.22549	-0.44755	0.20406
4	-0.03176	-0.16434	0.27015	-0.43682	0.21114
5	-0.04125	-0.20564	0.23931	-0.44510	0.20790
6	-0.06756	-0.19850	0.22936	-0.42717	0.22111
7	-0.01605	-0.14931	0.25430	-0.40242	0.23935
8	-0.03554	-0.20244	0.24000	-0.44053	0.21143

Population 1 through 8 are confirmation_checkerboard_01 through confirmation_checkerboard_08. C1 through C5 refer to the comparisons above. Both seed differences for every population remain in analysis/official_results/stage_B_v01/primary_analysis.json and the independent replay.

Spiral primary comparisons

Differences are first arm minus second arm common clipped BCE; negative values favor the first. Each interval uses eight population differences after averaging the two paired seeds. The nominal 99.75 percent intervals apply the registered Bonferroni family of 20 comparisons with 7 degrees of freedom.

Comparison	Mean	Interval	Interpretation
C1 MultiK versus ScalarK	-0.07797	[-0.12650, -0.02944]	First lower Beyond 0.01
C2 MultiK versus MultiHash	-0.29966	[-0.40834, -0.19097]	First lower Beyond 0.01
C3 MultiK versus TrigK	0.12199	[0.07796, 0.16601]	Second lower Beyond 0.01
C4 TrigK versus TrigHash	-0.41465	[-0.49024, -0.33905]	First lower Beyond 0.01
C5 TrigK versus TrigG121	0.17090	[0.09880, 0.24300]	Second lower Beyond 0.01

Beyond 0.01 means the entire interval exceeds the predeclared practical reference in the stated direction. Crossing zero is inconclusive, not evidence of equivalence. Small sample interval assumptions remain unverified.

All population differences

Population	C1	C2	C3	C4	C5
1	-0.07217	-0.32308	0.13383	-0.44237	0.14153
2	-0.12009	-0.34672	0.09559	-0.44063	0.13443
3	-0.07908	-0.34288	0.10295	-0.44396	0.14434
4	-0.01751	-0.16946	0.16751	-0.32936	0.24064
5	-0.09159	-0.33832	0.08762	-0.41845	0.17576
6	-0.06163	-0.25507	0.13760	-0.38137	0.21208
7	-0.09103	-0.36572	0.11088	-0.47500	0.11429
8	-0.09065	-0.25601	0.13992	-0.38603	0.20415

Population 1 through 8 are confirmation_spiral_01 through confirmation_spiral_08. C1 through C5 refer to the comparisons above. Both seed differences for every population remain in analysis/official_results/stage_B_v01/primary_analysis.json and the independent replay.

Noisy annulus primary comparisons

Differences are first arm minus second arm common clipped BCE; negative values favor the first. Each interval uses eight population differences after averaging the two paired seeds. The nominal 99.75 percent intervals apply the registered Bonferroni family of 20 comparisons with 7 degrees of freedom.

Comparison	Mean	Interval	Interpretation
C1 MultiK versus ScalarK	-0.00705	[-0.01160, -0.00250]	First lower 0.01 not established
C2 MultiK versus MultiHash	-0.11454	[-0.14173, -0.08735]	First lower Beyond 0.01
C3 MultiK versus TrigK	0.00527	[-0.00132, 0.01187]	Inconclusive 0.01 not established
C4 TrigK versus TrigHash	-0.07233	[-0.08954, -0.05512]	First lower Beyond 0.01
C5 TrigK versus TrigG121	-0.00057	[-0.01020, 0.00906]	Inconclusive 0.01 not established

Beyond 0.01 means the entire interval exceeds the predeclared practical reference in the stated direction. Crossing zero is inconclusive, not evidence of equivalence. Small sample interval assumptions remain unverified.

All population differences

Population	C1	C2	C3	C4	C5
1	-0.00843	-0.09517	0.00290	-0.05619	0.00135
2	-0.00775	-0.10932	0.00559	-0.06896	0.00781
3	-0.01082	-0.11922	0.01090	-0.08063	-0.00576
4	-0.00836	-0.11777	0.00907	-0.07122	-0.00916
5	-0.00722	-0.12904	0.00508	-0.07752	0.00657
6	-0.00792	-0.11070	-0.00199	-0.06089	-0.00389
7	-0.00403	-0.09224	0.00763	-0.07405	0.00141
8	-0.00188	-0.14289	0.00301	-0.08915	-0.00288

Population 1 through 8 are confirmation_annulus_01 through confirmation_annulus_08. C1 through C5 refer to the comparisons above. Both seed differences for every population remain in analysis/official_results/stage_B_v01/primary_analysis.json and the independent replay.

Rare disks primary comparisons

Differences are first arm minus second arm common clipped BCE; negative values favor the first. Each interval uses eight population differences after averaging the two paired seeds. The nominal 99.75 percent intervals apply the registered Bonferroni family of 20 comparisons with 7 degrees of freedom.

Comparison	Mean	Interval	Interpretation
C1 MultiK versus ScalarK	-0.00199	[-0.00596, 0.00198]	Inconclusive 0.01 not established
C2 MultiK versus MultiHash	-0.06910	[-0.07921, -0.05899]	First lower Beyond 0.01
C3 MultiK versus TrigK	0.00247	[-0.00083, 0.00577]	Inconclusive 0.01 not established
C4 TrigK versus TrigHash	-0.01207	[-0.01652, -0.00761]	First lower 0.01 not established
C5 TrigK versus TrigG121	0.00487	[0.00035, 0.00940]	Second lower 0.01 not established

Beyond 0.01 means the entire interval exceeds the predeclared practical reference in the stated direction. Crossing zero is inconclusive, not evidence of equivalence. Small sample interval assumptions remain unverified.

All population differences

Population	C1	C2	C3	C4	C5
1	-0.00039	-0.07232	-0.00068	-0.01169	0.00443
2	-0.00379	-0.06486	0.00127	-0.01117	0.00883
3	-0.00246	-0.06900	0.00247	-0.01171	0.00444
4	-0.00536	-0.07146	0.00455	-0.01519	0.00659
5	0.00030	-0.05548	0.00231	-0.00863	0.00051
6	-0.00238	-0.07256	0.00563	-0.01340	0.00723
7	0.00197	-0.07385	0.00090	-0.00863	0.00535
8	-0.00383	-0.07325	0.00332	-0.01612	0.00162

Population 1 through 8 are confirmation_disks_01 through confirmation_disks_08. C1 through C5 refer to the comparisons above. Both seed differences for every population remain in analysis/official_results/stage_B_v01/primary_analysis.json and the independent replay.

Validation milestone observations

Each entry is reached count / censored count / median updates among reached endpoints. There are 16 endpoints per task and arm. A dash means no endpoint reached the threshold. Observations are spaced 75 updates apart; every fit still completed all 15000 updates.

Checkerboard

Arm	90 percent	95 percent	97 percent
ScalarK	0 / 16 / —	0 / 16 / —	0 / 16 / —
MultiK	0 / 16 / —	0 / 16 / —	0 / 16 / —
MultiHash	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigK	10 / 6 / 9,862.5	0 / 16 / —	0 / 16 / —
TrigHash	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigG25	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigG121	16 / 0 / 225	16 / 0 / 712.5	16 / 0 / 2,287.5
MLPG129	0 / 16 / —	0 / 16 / —	0 / 16 / —

Spiral

Arm	90 percent	95 percent	97 percent
ScalarK	0 / 16 / —	0 / 16 / —	0 / 16 / —
MultiK	2 / 14 / 12,750	0 / 16 / —	0 / 16 / —
MultiHash	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigK	16 / 0 / 1,800	0 / 16 / —	0 / 16 / —
TrigHash	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigG25	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigG121	16 / 0 / 75	16 / 0 / 487.5	9 / 7 / 5,850
MLPG129	0 / 16 / —	0 / 16 / —	0 / 16 / —

These first observed validation crossings are descriptive learning diagnostics. The reached only median excludes censored endpoints and is not an overall time to target estimate. It is distinct from the test score of the checkpoint selected by validation BCE.

Complete reached update lists and nonreacher counts: analysis/official_results/stage_B_v01/descriptive_summary.json

Validation milestone observations

Each entry is reached count / censored count / median updates among reached endpoints. There are 16 endpoints per task and arm. A dash means no endpoint reached the threshold. Observations are spaced 75 updates apart; every fit still completed all 15000 updates.

Noisy annulus

Arm	90 percent	95 percent	97 percent
ScalarK	0 / 16 / —	0 / 16 / —	0 / 16 / —
MultiK	0 / 16 / —	0 / 16 / —	0 / 16 / —
MultiHash	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigK	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigHash	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigG25	0 / 16 / —	0 / 16 / —	0 / 16 / —
TrigG121	0 / 16 / —	0 / 16 / —	0 / 16 / —
MLPG129	0 / 16 / —	0 / 16 / —	0 / 16 / —

Rare disks

Arm	90 percent	95 percent	97 percent
ScalarK	16 / 0 / 525	16 / 0 / 787.5	16 / 0 / 1,087.5
MultiK	16 / 0 / 375	16 / 0 / 675	16 / 0 / 1,162.5
MultiHash	15 / 1 / 3,975	0 / 16 / —	0 / 16 / —
TrigK	16 / 0 / 75	16 / 0 / 150	16 / 0 / 300
TrigHash	16 / 0 / 150	16 / 0 / 375	14 / 2 / 1,125
TrigG25	16 / 0 / 150	16 / 0 / 375	16 / 0 / 600
TrigG121	16 / 0 / 75	16 / 0 / 150	16 / 0 / 150
MLPG129	13 / 3 / 2,850	8 / 8 / 8,850	4 / 12 / 10,350

These first observed validation crossings are descriptive learning diagnostics. The reached only median excludes censored endpoints and is not an overall time to target estimate. It is distinct from the test score of the checkpoint selected by validation BCE.

Complete reached update lists and nonreacher counts: analysis/official_results/stage_B_v01/descriptive_summary.json

Checkerboard geometric margin diagnostics

The same train fitted geometric margin bins are applied to every arm. Margin is the gap between the second nearest and nearest squared centroid distances. Q1 is nearest a routing boundary; Q4 has the largest gap. This is not distance to the true class boundary.

Arm	Q1	Q2	Q3	Q4
ScalarK	0.651 (0.078) 67.16 (5.98)%	0.465 (0.026) 77.83 (2.06)%	0.467 (0.036) 80.66 (3.31)%	0.600 (0.012) 66.83 (1.95)%
MultiK	0.615 (0.128) 68.20 (5.42)%	0.458 (0.036) 77.81 (1.27)%	0.444 (0.028) 82.87 (1.46)%	0.535 (0.030) 74.54 (4.32)%
MultiHash	0.703 (0.014) 46.90 (5.87)%	0.703 (0.010) 48.05 (4.90)%	0.700 (0.005) 50.51 (3.26)%	0.696 (0.003) 50.72 (1.65)%
TrigK	0.339 (0.103) 84.33 (4.97)%	0.294 (0.034) 87.42 (1.60)%	0.228 (0.022) 92.10 (2.27)%	0.260 (0.015) 91.26 (1.24)%
TrigHash	0.697 (0.010) 48.69 (3.98)%	0.698 (0.006) 49.33 (3.49)%	0.697 (0.007) 50.20 (3.79)%	0.699 (0.003) 49.05 (1.87)%
TrigG25	0.695 (0.008) 50.54 (7.47)%	0.695 (0.008) 48.62 (7.62)%	0.699 (0.007) 47.23 (5.52)%	0.694 (0.003) 53.10 (2.79)%
TrigG121	0.050 (0.009) 98.32 (0.53)%	0.063 (0.014) 97.88 (1.00)%	0.038 (0.009) 99.02 (0.50)%	0.051 (0.006) 98.40 (0.42)%
MLPG129	0.618 (0.032) 61.83 (5.12)%	0.607 (0.033) 64.10 (5.09)%	0.606 (0.028) 64.24 (3.80)%	0.588 (0.027) 68.08 (2.95)%

Each cell shows common BCE mean (population SD), then macro F1 percent mean (population SD). The two seeds are averaged within each observed population. These are descriptive conditional summaries, not additional primary comparisons.

Shared records and class composition

Bin	Test n range	Positive class %	Observed	Empty	Absent
Q1	152–190	52.57 (5.51)	8	0	0
Q2	218–273	50.74 (2.22)	8	0	0
Q3	378–430	48.23 (2.44)	8	0	0
Q4	752–835	49.85 (1.33)	8	0	0

Counts refer to unique test records per population, not duplicated seeds. Observed, empty and absent count population bins. Train fitted boundaries vary across populations, so bin labels do not denote identical coordinate regions.

Complete population counts, class proportions and all hash/geometric bank slices: analysis/official_results/stage_B_v01/diagnostic_summary.json

Spiral geometric margin diagnostics

The same train fitted geometric margin bins are applied to every arm. Margin is the gap between the second nearest and nearest squared centroid distances. Q1 is nearest a routing boundary; Q4 has the largest gap. This is not distance to the true class boundary.

Arm	Q1	Q2	Q3	Q4
ScalarK	0.617 (0.096) 79.42 (6.09)%	0.319 (0.127) 88.72 (5.65)%	0.414 (0.026) 81.39 (2.89)%	0.470 (0.028) 78.45 (2.44)%
MultiK	0.648 (0.082) 82.62 (6.21)%	0.273 (0.132) 89.83 (5.94)%	0.355 (0.045) 84.19 (2.50)%	0.349 (0.062) 88.09 (3.80)%
MultiHash	0.688 (0.016) 50.45 (3.51)%	0.704 (0.021) 49.20 (4.49)%	0.662 (0.011) 56.39 (2.39)%	0.662 (0.010) 60.45 (2.68)%
TrigK	0.463 (0.120) 88.78 (2.52)%	0.203 (0.098) 94.42 (3.47)%	0.253 (0.038) 90.40 (2.23)%	0.214 (0.039) 93.58 (1.23)%
TrigHash	0.677 (0.020) 54.39 (4.54)%	0.702 (0.020) 49.94 (4.33)%	0.662 (0.013) 57.01 (3.52)%	0.651 (0.005) 61.07 (1.68)%
TrigG25	0.674 (0.015) 54.71 (3.12)%	0.698 (0.029) 48.30 (4.09)%	0.650 (0.012) 58.98 (3.99)%	0.640 (0.005) 62.59 (1.47)%
TrigG121	0.045 (0.018) 98.12 (1.53)%	0.054 (0.020) 97.28 (1.45)%	0.063 (0.020) 97.10 (1.44)%	0.101 (0.013) 95.75 (0.80)%
MLPG129	0.543 (0.035) 70.83 (5.20)%	0.547 (0.037) 70.71 (4.99)%	0.556 (0.049) 67.59 (4.11)%	0.483 (0.056) 73.93 (5.93)%

Each cell shows common BCE mean (population SD), then macro F1 percent mean (population SD). The two seeds are averaged within each observed population. These are descriptive conditional summaries, not additional primary comparisons.

Shared records and class composition

Bin	Test n range	Positive class %	Observed	Empty	Absent
Q1	157–193	50.74 (3.83)	8	0	0
Q2	224–273	50.48 (2.85)	8	0	0
Q3	377–408	48.95 (3.52)	8	0	0
Q4	767–824	50.26 (1.16)	8	0	0

Counts refer to unique test records per population, not duplicated seeds. Observed, empty and absent count population bins. Train fitted boundaries vary across populations, so bin labels do not denote identical coordinate regions.

Complete population counts, class proportions and all hash/geometric bank slices: analysis/official_results/stage_B_v01/diagnostic_summary.json

Noisy annulus geometric margin diagnostics

The same train fitted geometric margin bins are applied to every arm. Margin is the gap between the second nearest and nearest squared centroid distances. Q1 is nearest a routing boundary; Q4 has the largest gap. This is not distance to the true class boundary.

Arm	Q1	Q2	Q3	Q4
ScalarK	0.481 (0.073) 79.17 (4.37)%	0.420 (0.034) 84.64 (1.37)%	0.390 (0.037) 86.97 (1.03)%	0.395 (0.020) 87.16 (1.43)%
MultiK	0.448 (0.075) 83.02 (4.55)%	0.409 (0.037) 85.97 (2.13)%	0.387 (0.039) 86.81 (2.08)%	0.392 (0.023) 87.48 (1.49)%
MultiHash	0.599 (0.043) 70.47 (5.37)%	0.526 (0.024) 75.78 (2.71)%	0.482 (0.049) 80.35 (3.38)%	0.509 (0.020) 78.20 (0.96)%
TrigK	0.437 (0.084) 83.76 (4.35)%	0.408 (0.037) 86.60 (1.98)%	0.386 (0.037) 87.59 (1.72)%	0.385 (0.024) 88.01 (1.44)%
TrigHash	0.574 (0.052) 73.60 (4.99)%	0.507 (0.027) 76.37 (2.56)%	0.460 (0.049) 81.77 (1.71)%	0.434 (0.016) 83.45 (1.46)%
TrigG25	0.544 (0.056) 76.78 (4.89)%	0.492 (0.027) 77.05 (3.07)%	0.453 (0.046) 82.48 (1.90)%	0.424 (0.017) 84.73 (1.54)%
TrigG121	0.436 (0.091) 84.36 (4.70)%	0.419 (0.034) 85.28 (1.21)%	0.394 (0.036) 87.07 (1.86)%	0.378 (0.026) 87.68 (1.23)%
MLPG129	0.461 (0.071) 83.54 (4.75)%	0.432 (0.026) 85.38 (1.39)%	0.394 (0.040) 87.04 (1.51)%	0.435 (0.031) 86.01 (1.58)%

Each cell shows common BCE mean (population SD), then macro F1 percent mean (population SD). The two seeds are averaged within each observed population. These are descriptive conditional summaries, not additional primary comparisons.

Shared records and class composition

Bin	Test n range	Positive class %	Observed	Empty	Absent
Q1	141–186	44.25 (5.93)	8	0	0
Q2	219–256	50.28 (3.09)	8	0	0
Q3	380–431	60.18 (4.05)	8	0	0
Q4	744–820	40.20 (1.92)	8	0	0

Counts refer to unique test records per population, not duplicated seeds. Observed, empty and absent count population bins. Train fitted boundaries vary across populations, so bin labels do not denote identical coordinate regions.

Complete population counts, class proportions and all hash/geometric bank slices: analysis/official_results/stage_B_v01/diagnostic_summary.json

Rare disks geometric margin diagnostics

The same train fitted geometric margin bins are applied to every arm. Margin is the gap between the second nearest and nearest squared centroid distances. Q1 is nearest a routing boundary; Q4 has the largest gap. This is not distance to the true class boundary.

Arm	Q1	Q2	Q3	Q4
ScalarK	0.026 (0.018) 86.51 (22.55)%	0.025 (0.019) 95.33 (4.96)%	0.023 (0.008) 98.56 (0.86)%	0.011 (0.003) 99.22 (0.70)%
MultiK	0.021 (0.017) 85.95 (22.23)%	0.021 (0.014) 95.35 (5.83)%	0.020 (0.006) 98.75 (0.62)%	0.011 (0.004) 99.05 (0.55)%
MultiHash	0.092 (0.055) 81.05 (19.27)%	0.089 (0.039) 86.55 (13.68)%	0.097 (0.018) 90.87 (1.93)%	0.076 (0.024) 91.30 (3.41)%
TrigK	0.017 (0.016) 86.39 (22.49)%	0.014 (0.009) 96.86 (5.59)%	0.016 (0.005) 99.10 (0.81)%	0.010 (0.005) 99.01 (0.60)%
TrigHash	0.023 (0.017) 86.19 (22.35)%	0.029 (0.016) 94.89 (5.64)%	0.036 (0.012) 96.99 (0.87)%	0.019 (0.004) 98.02 (1.02)%
TrigG25	0.016 (0.013) 86.50 (22.54)%	0.018 (0.010) 98.15 (3.42)%	0.022 (0.006) 99.09 (0.69)%	0.013 (0.005) 99.05 (0.50)%
TrigG121	0.010 (0.016) 86.67 (22.68)%	0.010 (0.005) 98.09 (3.39)%	0.008 (0.006) 99.50 (0.63)%	0.007 (0.003) 99.11 (0.71)%
MLPG129	0.098 (0.053) 80.36 (19.10)%	0.096 (0.044) 85.26 (15.25)%	0.095 (0.038) 89.34 (5.42)%	0.053 (0.033) 93.16 (4.97)%

Each cell shows common BCE mean (population SD), then macro F1 percent mean (population SD). The two seeds are averaged within each observed population. These are descriptive conditional summaries, not additional primary comparisons.

In Q1, populations 01 and 07 contain 190 and 168 test records, respectively, with no positives. All 32 arm and seed prediction sets classify these records correctly, yet macro F1 is 50 percent because the absent positive class contributes zero. This is the slice's maximum under the fixed two-class convention. A lower Q1 mean and larger SD alone therefore do not establish routing boundary errors; read BCE and class composition alongside F1.

Shared records and class composition

Bin	Test n range	Positive class %	Observed	Empty	Absent
Q1	141–190	13.84 (9.34)	8	0	0
Q2	219–254	12.57 (7.19)	8	0	0
Q3	368–443	12.97 (2.72)	8	0	0
Q4	766–832	10.09 (4.49)	8	0	0

Counts refer to unique test records per population, not duplicated seeds. Observed, empty and absent count population bins. Train fitted boundaries vary across populations, so bin labels do not denote identical coordinate regions.

Bank exposure and momentum diagnostics

Within each population, hash cutpoints match the geometric banks' training occupancies exactly. Heldout occupancy and class composition remain free to differ. K denotes geometric routing and H hash routing. The following ranges include all five banks across eight populations; banks are not matched semantic groups.

Task route	Training n	Test n	Test positive %
Checkerboard K	739–1217	227–403	43.61–54.13
Checkerboard H	739–1217	249–387	42.30–56.00
Spiral K	767–1226	244–402	42.36–57.32
Spiral H	767–1226	243–420	45.27–54.21
Noisy annulus K	751–1173	247–406	9.69–60.32
Noisy annulus H	751–1173	211–425	42.18–51.18
Rare disks K	720–1183	238–407	0.00–21.80
Rare disks H	720–1183	236–417	8.91–13.98

Complete population counts, class proportions and all hash/geometric bank slices: analysis/official_results/stage_B_v01/diagnostic_summary.json

Absent bank updates and retained movement

Arm	Checkerboard	Spiral	Annulus	Disks
ScalarK	A 2 M 0.00936	A 1 M 0.00906	A 1 M 0.00515	A 2 M 0.01217
MultiK	A 2 M 0.00961	A 1 M 0.01208	A 1 M 0.00615	A 2 M 0.01558
MultiHash	A 2 M 0.01040	A 1 M 0.01000	A 3 M 0.01628	A 1 M 0.00925
TrigK	A 2 M 0.01878	A 1 M 0.01032	A 1 M 0.01181	A 2 M 0.01718
TrigHash	A 2 M 0.01486	A 1 M 0.01029	A 3 M 0.02329	A 1 M 0.00999
TrigG25	A 0 M 0.00000	A 0 M 0.00000	A 0 M 0.00000	A 0 M 0.00000
TrigG121	A 0 M 0.00000	A 0 M 0.00000	A 0 M 0.00000	A 0 M 0.00000
MLPG129	A 0 M 0.00000	A 0 M 0.00000	A 0 M 0.00000	A 0 M 0.00000

A is the maximum count of updates with no records for that bank. M is the maximum cumulative parameter movement during such updates, across the 16 fits and their banks. Nonzero movement is consistent with retained Adam moments; it is not fresh data exposure. Global arms have one bank and no absent updates.

Each training record is visited in every one of 200 rounds. Per bank visits, minibatch appearances, absence counts and movement are retained in each committed metrics.json under training_mechanism_and_costs.

Complete evidence

All selected heads, coefficient vectors, parameter norms, exposure counters and costs: analysis/official_results/stage_B_v01/endpoints/B/<population>/seed_<1401 or 1402>/<arm>/metrics.json. The exact 512 paths and hashes are listed in analysis/official_results/stage_B_v01/endpoint_index.json.

All population mean values, intervals, differences, milestones and detailed diagnostic blocks: report_work/final_report/measured_capsule_v01/MEASURED_REPORT_CAPSULE.json. Independent replay: analysis/independent_results/stage_B_v01/REPLAY_COMMIT.json.

The four task families remain separate throughout. These results support local conclusions about routing, readout and basis capacity on small synthetic problems. They do not validate the full Q Pathformer architecture, language training, external task performance or physical quantum execution.

Parameter and readout diagnostics

P is the mean selected bank parameter L2 norm. H is the largest absolute recorded output coefficient across all selected fits and banks. G is the mean cumulative bank gradient L2 sum over the full training horizon. These magnitudes describe optimization; their different parameter scales prevent a capacity or causal performance interpretation.

Arm	Checkerboard	Spiral	Annulus	Disks
ScalarK	P 16.40 H 47.01 G 3999.7	P 21.72 H 60.63 G 4361.3	P 6.10 H 8.65 G 1440.2	P 24.54 H 30.51 G 932.9
MultiK	P 17.04 H 31.35 G 3782.7	P 24.58 H 50.96 G 3783.3	P 6.48 H 8.18 G 1592.2	P 22.93 H 29.80 G 846.5
MultiHash	P 1.77 H 1.42 G 641.5	P 3.21 H 2.23 G 823.7	P 6.60 H 9.08 G 1360.0	P 21.71 H 19.64 G 1353.6
TrigK	P 26.39 H 17.05 G 738.4	P 24.01 H 13.93 G 606.3	P 4.16 H 2.99 G 773.4	P 20.42 H 14.88 G 111.0
TrigHash	P 0.33 H 0.22 G 1218.8	P 1.12 H 0.85 G 1170.1	P 3.87 H 3.56 G 869.9	P 22.56 H 14.79 G 171.3
TrigG25	P 0.26 H 0.19 G 2786.8	P 1.18 H 0.72 G 2681.0	P 3.89 H 3.02 G 2026.6	P 28.49 H 17.77 G 473.1
TrigG121	P 43.05 H 32.20 G 1565.3	P 33.34 H 14.58 G 1581.1	P 3.93 H 2.67 G 3666.5	P 28.91 H 13.99 G 460.3
MLPG129	P 19.26 H 17.28 G 6203.7	P 41.25 H 30.49 G 6568.0	P 23.75 H 7.27 G 7705.4	P 36.70 H 21.02 G 3930.6

P and G summarize 80 bank observations per routed arm and task, or 16 for a global arm. These are nested descriptive observations, not additional independent samples. H uses saved scalar/flexible readout coefficients for circuits, complete coefficients for Trig, and complete weights for the MLP.

The selected values come from the earliest checkpoint minimizing validation BCE after full training. They were never chosen by test performance. Large coefficients can amplify measurement error; that hypothesis must be checked against the separate joint shot results before a hardware recommendation.

Conversion preserves the selected function

All 192 selected circuit models were converted to real trigonometric logits. A separate implementation reconstructed the coefficients through Laurent algebra and explicit dense-state calculations. It also checked 256 learned Trig models repacked for the common feature kernel. These are 448 representations of existing checkpoints, not additional trained models.

Each circuit bank uses 25 coefficients: the tensor product of [1, cos(pi x), sin(pi x), cos(2 pi x), sin(2 pi x)] on the two coordinates. The affine readout is absorbed into the logit; the frozen router and sigmoid remain. Two input-rotation occurrences per coordinate give the registered frequency support. The producer uses discrete orthogonal projection at 25 fixed, label-free nodes per bank; the independent reconstruction uses separate Laurent coefficients.

Comparison: all checked splits/fixtures	Max probability difference	Max logit difference
Producer: original vs converted circuit	9.936e-15	7.816e-14
Producer: original vs repacked learned Trig	6.772e-15	5.684e-14
Independent replay: all representations	1.893e-14	9.059e-14

All class decisions agreed. Independent circuit coefficient error was at most 1.776e-14. Validation/test checks cover 1,433,600 probability rows; off-grid, routed boundary and forced-bank fixtures are additional checks, not independent training samples.

Circuit arm	Original parameters B	Coefficient table B	Retain both B
ScalarK	400	1000	1400
MultiK	560	1000	1560
MultiHash	560	1000	1560

Each five-bank circuit table contains 125 float64 coefficients, or 1,000 parameter bytes. These totals exclude routers, file containers and retained construction evidence. Conversion is not compression and confers no accuracy gain. Its algebraic equivalence is specific to the tested two-qubit, two-layer core and linear readout; it does not establish an efficient shortcut for arbitrary quantum states, deeper circuits or the complete Q-Pathformer architecture.

Construction loading and verification costs

The following values summarize one instrumented construction and one checkpoint-loading operation for each of 64 selected models per arm. They describe this pass, not isolated repeated latency benchmarks or independent replications. Loading includes model construction and selected-checkpoint loading; compilation/repacking excludes file serialization and the later parity checks.

Arm	Median construction ms	Median load ms	Median retained NPZ B
ScalarK	1.9485	1.568	6,588
MultiK	1.9174	1.574	10,588
MultiHash	1.9886	1.574	10,588
TrigK	0.0395	1.434	1,278
TrigHash	0.0409	1.442	1,278
TrigG25	0.0380	1.476	478
TrigG121	0.0392	1.407	1,246

Circuit NPZ files retain construction nodes/moments/logits as evidence as well as coefficients. Their file sizes are not bare deployment parameter sizes. Learned Trig repacking changes representation only; basis widths remain 25 or 121 as registered.

Phase	Whole invocation wall s	Inner wall s	Inner CPU s
Producer conversion	183.79	18.29	12.91
Independent conversion audit	286.20	113.34	109.77

The producer recorded 174.48s process CPU through its resource sample, including imports and release verification. The independent audit consumed 279.59s process CPU through exit. Inner timers begin after the release/hash gate; they are nested within whole-invocation time and must not be added to it. Differences include work outside the inner timer, not a separately measured gate-only cost.

These checks establish function preservation and the costs of constructing and verifying the representation. They do not establish a throughput gain. The separately audited, three-session inference benchmark supplies that comparison; physical quantum-device loading, noise and runtime are outside this classical conversion result.

Sources: completed conversion commit and per-checkpoint records; completed independent conversion commit/replay; producer and independent invocation cost records; frozen conversion protocol and implementations.

Checkerboard inference latency

Microseconds per complete raw input call, including routing, features or circuit core, head and sigmoid. Take the median of 15 trials, then the median of three session medians per model; average paired seeds within each population. Tables show medians across eight populations.

Model	Batch 1 µs	Batch 32 µs	Batch 1024 µs
ScalarK original	294.57	1,380.45	2,104.70
ScalarK compiled	68.38	93.25	436.10
MultiK original	316.55	1,481.40	2,255.65
MultiK compiled	69.60	95.83	431.98
MultiHash original	316.82	1,529.60	3,277.15
MultiHash compiled	74.75	135.55	1,378.78
TrigK	61.25	69.45	348.38
TrigHash	64.35	101.12	1,294.78
TrigG25	30.82	36.05	189.62
TrigG121	43.40	54.40	929.77
MLPG129	21.10	23.95	80.62

Batch latencies include every record. Learned Trig controls use the common optimized kernel. Compiled circuit rows preserve the selected predictor.

Matched conversion savings

Arm	Batch 1	Batch 32	Batch 1024
ScalarK	4.31x +222.3µs BE 5.6–16.5	14.70x +1269.8µs BE 1.1–3.0	4.87x +1654.8µs BE 0.8–2.4
MultiK	4.58x +241.2µs BE 4.7–20.2	15.41x +1376.7µs BE 1.1–3.7	5.24x +1796.1µs BE 0.8–2.9
MultiHash	4.31x +238.6µs BE 5.7–15.9	12.14x +1388.5µs BE 0.9–2.9	2.32x +1824.8µs BE 0.7–2.3

Ratios and savings are computed on matched trials before aggregation; they are not division or subtraction of the displayed aggregate latency cells. BE is construction only break even in complete-call equivalents at that batch size; round upward to obtain a whole number of calls. The range covers all 48 model/session records per cell; 432 of 432 records have positive savings. Loading, storage and hardware costs are excluded; technical repeats do not increase independent sample size.

Spiral inference latency

Microseconds per complete raw input call, including routing, features or circuit core, head and sigmoid. Take the median of 15 trials, then the median of three session medians per model; average paired seeds within each population. Tables show medians across eight populations.

Model	Batch 1 µs	Batch 32 µs	Batch 1024 µs
ScalarK original	299.12	1,412.58	2,098.07
ScalarK compiled	69.35	95.58	440.35
MultiK original	316.68	1,511.58	2,265.45
MultiK compiled	69.15	99.00	438.82
MultiHash original	324.30	1,518.42	3,330.32
MultiHash compiled	74.55	130.60	1,371.75
TrigK	60.35	69.12	348.65
TrigHash	65.12	99.75	1,296.10
TrigG25	30.93	35.67	190.80
TrigG121	43.23	53.92	925.88
MLPG129	21.05	24.00	80.88

Batch latencies include every record. Learned Trig controls use the common optimized kernel. Compiled circuit rows preserve the selected predictor.

Matched conversion savings

Arm	Batch 1	Batch 32	Batch 1024
ScalarK	4.28x +224.4µs BE 6.6–21.6	14.79x +1296.3µs BE 0.9–3.9	4.95x +1640.1µs BE 0.8–3.1
MultiK	4.55x +242.9µs BE 6.0–19.9	15.70x +1414.4µs BE 1.1–3.7	5.21x +1780.0µs BE 0.7–2.9
MultiHash	4.36x +243.8µs BE 4.9–23.0	11.90x +1382.3µs BE 1.0–4.1	2.33x +1888.6µs BE 0.8–3.2

Ratios and savings are computed on matched trials before aggregation; they are not division or subtraction of the displayed aggregate latency cells. BE is construction only break even in complete-call equivalents at that batch size; round upward to obtain a whole number of calls. The range covers all 48 model/session records per cell; 432 of 432 records have positive savings. Loading, storage and hardware costs are excluded; technical repeats do not increase independent sample size.

Noisy annulus inference latency

Microseconds per complete raw input call, including routing, features or circuit core, head and sigmoid. Take the median of 15 trials, then the median of three session medians per model; average paired seeds within each population. Tables show medians across eight populations.

Model	Batch 1 µs	Batch 32 µs	Batch 1024 µs
ScalarK original	305.77	1,384.25	2,124.03
ScalarK compiled	70.70	99.80	432.62
MultiK original	317.18	1,470.72	2,253.95
MultiK compiled	70.53	95.83	445.48
MultiHash original	319.15	1,502.67	3,332.03
MultiHash compiled	73.80	127.78	1,387.50
TrigK	61.98	70.75	347.48
TrigHash	64.92	100.40	1,316.83
TrigG25	31.73	35.65	188.82
TrigG121	43.55	54.60	949.73
MLPG129	21.00	23.88	81.25

Batch latencies include every record. Learned Trig controls use the common optimized kernel. Compiled circuit rows preserve the selected predictor.

Matched conversion savings

Arm	Batch 1	Batch 32	Batch 1024
ScalarK	4.32x +226.0µs BE 5.9–17.2	14.55x +1264.2µs BE 0.9–3.1	4.92x +1653.5µs BE 0.8–2.5
MultiK	4.55x +240.8µs BE 3.9–17.1	15.97x +1353.6µs BE 0.7–3.2	5.15x +1788.8µs BE 0.5–2.5
MultiHash	4.36x +245.2µs BE 5.7–19.6	12.03x +1357.2µs BE 1.0–3.6	2.36x +1919.3µs BE 0.7–2.8

Ratios and savings are computed on matched trials before aggregation; they are not division or subtraction of the displayed aggregate latency cells. BE is construction only break even in complete-call equivalents at that batch size; round upward to obtain a whole number of calls. The range covers all 48 model/session records per cell; 432 of 432 records have positive savings. Loading, storage and hardware costs are excluded; technical repeats do not increase independent sample size.

Rare disks inference latency

Microseconds per complete raw input call, including routing, features or circuit core, head and sigmoid. Take the median of 15 trials, then the median of three session medians per model; average paired seeds within each population. Tables show medians across eight populations.

Model	Batch 1 µs	Batch 32 µs	Batch 1024 µs
ScalarK original	300.12	1,388.40	2,116.47
ScalarK compiled	69.53	93.45	425.65
MultiK original	317.98	1,494.38	2,242.68
MultiK compiled	71.10	101.47	423.18
MultiHash original	322.10	1,525.35	3,256.22
MultiHash compiled	72.90	134.35	1,363.38
TrigK	60.98	68.85	346.95
TrigHash	65.58	101.20	1,286.40
TrigG25	31.25	35.80	185.82
TrigG121	43.00	54.27	925.50
MLPG129	20.85	23.98	82.10

Batch latencies include every record. Learned Trig controls use the common optimized kernel. Compiled circuit rows preserve the selected predictor.

Matched conversion savings

Arm	Batch 1	Batch 32	Batch 1024
ScalarK	4.29x +228.2µs BE 4.8–22.0	14.59x +1279.8µs BE 1.0–4.0	5.06x +1652.1µs BE 0.6–3.2
MultiK	4.58x +243.6µs BE 6.0–19.9	15.77x +1394.0µs BE 0.8–3.6	5.38x +1817.6µs BE 0.5–2.9
MultiHash	4.34x +243.9µs BE 4.5–21.3	11.78x +1380.7µs BE 0.7–3.9	2.34x +1824.6µs BE 0.6–3.0

Ratios and savings are computed on matched trials before aggregation; they are not division or subtraction of the displayed aggregate latency cells. BE is construction only break even in complete-call equivalents at that batch size; round upward to obtain a whole number of calls. The range covers all 48 model/session records per cell; 432 of 432 records have positive savings. Loading, storage and hardware costs are excluded; technical repeats do not increase independent sample size.

Timing variation across sessions

Session 3 was materially faster than sessions 1 and 2. CPU and operating-system conditions were not fully controlled, and the cause was not measured. All three sessions are retained. The following medians describe each task's 528 matched conditions; they are not independent replications or performance hypothesis tests.

Task	Session max/min	Session 3/1	Last five / first five trials in sessions 1 2 3
Checkerboard	1.76	0.592	0.975 / 0.971 / 0.986
Spiral	1.77	0.591	0.975 / 0.974 / 0.987
Noisy annulus	1.77	0.589	0.967 / 0.978 / 0.984
Rare disks	1.77	0.595	0.965 / 0.970 / 0.986

The full inner timing sections took 34.23, 33.37 and 18.94 seconds. Conversion ratios are paired within sessions before aggregation. Their stability does not remove absolute latency drift or support an estimate of physical quantum latency.

The independent audit checks all 95,040 archived timed calls, their ordering, pairing and summary statistics. The 12,672 warmups have counters only; individual warmup calls were not retained as raw timing records.

Detailed population values, p10/p90, ranges and session medians: analysis/independent_timing_results/timing_v01/cross_session_descriptive.json. Trial summaries: session1_replay.json through session3_replay.json in that directory.

Timing and verification costs

These costs describe running and verifying the timing exercise. They are separate from the per-call inference latencies on the preceding pages. Each timing session retained 31,680 timed calls across 2,112 conditions; the independent audit replayed all 95,040 archived trial records without running new models or timing calls.

Phase	Inner wall s	Inner CPU s	Process CPU s	Outer wall s
Timing session 1	34.23	32.45	255.33	260.16
Timing session 2	33.37	31.81	254.05	258.73
Timing session 3	18.94	17.92	234.02	237.76
Independent timing audit	4.24	4.20	98.45	98.95

The columns have nested scopes and must not be added together. Inner wall and CPU cover the instrumented timing or replay section. Process CPU includes imports, release verification and hash checks as well as the inner work. For timing sessions it is sampled before final process exit; for the independent audit it was read from the retained Windows leaf-process handle after exit. It excludes the shell and Python environment redirector. Outer wall spans the invocation, including startup and preflight. CPU measures processor time; wall measures elapsed time.

Session 3 had a shorter inner wall time than sessions 1 and 2, and all three remain in the descriptive estimator. The cause was not measured. Release verification and hash checks account for substantial work outside the inner sections; these governance costs are not a model's per-call deployment cost. No physical quantum execution or hardware latency is included.

Sources: analysis/deployment_results/timing_session1_v01 through timing_session3_v01/summary.json; governance/TIMING_SESSION1_COMPLETION_v01.json through TIMING_SESSION3_COMPLETION_v01.json. Independent audit: analysis/independent_timing_results/timing_v01/summary.json, timing_v01_invocation.json and timing_v01_process_cost.json. INFERENCE_BLOCKS_SOURCE_REVIEW_v02.json retains exact paths, hashes and unrounded costs.

Finite shot diagnostics total budget 1024

Larger measurement budgets reduced average sampling error in every task–arm group. These results measure variation around each trained model's own ideal prediction. They do not measure classification accuracy or hardware performance.

Each row contains 16 selected checkpoints: two training seeds within each of eight independent populations. Each checkpoint uses its 32 lowest numeric validation IDs, chosen without labels or prediction values, with 20 technical repeats. Record-level diagnostics are averaged within checkpoint, then across the two seeds, then equally across eight populations. All 12 task–arm groups are shown.

Task	Arm	MAE pp	MSE ×1e3	Bias pp	Bias² ×1e3	Var ×1e3	Flip %
Checkerboard	ScalarK	6.209	8.563	+0.095	0.582	7.981	11.172
Checkerboard	MultiK	7.923	13.098	-0.026	0.965	12.133	13.379
Checkerboard	MultiHash	0.763	0.095	-0.005	0.005	0.090	6.016
Spiral	ScalarK	6.516	12.256	+0.189	1.303	10.953	9.502
Spiral	MultiK	8.270	17.959	+0.398	1.933	16.026	10.039
Spiral	MultiHash	1.302	0.297	+0.013	0.015	0.282	5.566
Noisy annulus	ScalarK	1.530	0.549	+0.058	0.030	0.518	1.660
Noisy annulus	MultiK	2.388	1.418	+0.013	0.065	1.353	2.686
Noisy annulus	MultiHash	2.751	1.621	+0.003	0.096	1.525	4.189
Rare disks	ScalarK	0.420	0.646	+0.026	0.042	0.604	0.352
Rare disks	MultiK	0.466	0.926	-0.075	0.067	0.859	0.703
Rare disks	MultiHash	1.780	2.879	+0.226	0.196	2.683	1.201

MAE and signed bias are probability percentage points; MSE, squared bias and variance are multiplied by 1,000. Bias² is the mean squared per-record technical bias, not the square of the displayed signed mean. Var uses ddof=0 over 20 repeats. MSE = Bias² + Var before rounding. Flip is disagreement with the own-ideal p≥0.5 decision, not error against labels.

ScalarK uses all 1,024 shots in the XX setting. MultiK and MultiHash use 512 in XX and 512 in ZZ. The five moments are estimated jointly within those two settings; they are not five independent allocations of 1,024 shots.

Finite shot diagnostics total budget 8192

The same 192 checkpoints and fixed inputs were evaluated with eight times the total measurement budget. Mean probability MAE, MSE and decision disagreement decreased in all 12 task–arm groups. This is a descriptive comparison of sampling stability, not an additional primary hypothesis test or a claim about real devices.

Task	Arm	MAE pp	MSE ×1e3	Bias pp	Bias² ×1e3	Var ×1e3	Flip %
Checkerboard	ScalarK	2.285	1.212	+0.043	0.062	1.150	4.609
Checkerboard	MultiK	2.853	1.741	+0.026	0.078	1.663	5.215
Checkerboard	MultiHash	0.270	0.012	-0.001	0.001	0.012	2.002
Spiral	ScalarK	2.360	1.689	+0.057	0.097	1.591	3.984
Spiral	MultiK	2.970	2.376	-0.051	0.154	2.221	3.779
Spiral	MultiHash	0.467	0.038	-0.000	0.002	0.036	1.943
Noisy annulus	ScalarK	0.537	0.068	+0.002	0.003	0.065	0.488
Noisy annulus	MultiK	0.833	0.175	+0.000	0.009	0.167	0.859
Noisy annulus	MultiHash	0.956	0.194	-0.007	0.012	0.182	1.592
Rare disks	ScalarK	0.158	0.085	+0.012	0.004	0.081	0.146
Rare disks	MultiK	0.192	0.169	-0.003	0.010	0.160	0.508
Rare disks	MultiHash	0.603	0.316	+0.014	0.014	0.303	0.225

MAE and signed bias are probability percentage points; MSE, squared bias and variance are multiplied by 1,000. Bias² is the mean squared per-record technical bias, not the square of the displayed signed mean. Var uses ddof=0 over 20 repeats. MSE = Bias² + Var before rounding. Flip is disagreement with the own-ideal p≥0.5 decision, not error against labels.

At total budget 8,192, ScalarK uses 8,192 XX shots and the multi-moment arms use 4,096 XX plus 4,096 ZZ shots. Lower shot error does not establish a better learned classifier: a weak or less responsive ideal model can also be stable under sampling. Task performance must be assessed separately using the sealed test results.

The hierarchy remains eight independent populations per task. The paired seeds, 32 records and 20 repeats are nested observations; neither 16 checkpoints nor 10,240 probability values per row provides a larger independent population sample. These descriptive tables contain no confidence intervals or significance tests.

Head amplification joint sampling and costs

A learned output head can amplify measurement variation. The following table shows its mean absolute-weight sum and logit variances, averaged with the same population hierarchy. The analytic variance uses the full within-setting covariance; the observed variance uses 20 finite repeats and ddof=0.

Task	Arm	Head L1	Analytic 1,024	Observed 1,024	Analytic 8,192	Observed 8,192
Checkerboard	ScalarK	14.82	0.2994	0.2791	0.0374	0.0360
Checkerboard	MultiK	24.98	0.4442	0.4240	0.0555	0.0521
Checkerboard	MultiHash	1.33	0.0016	0.0015	0.0002	0.0002
Spiral	ScalarK	20.70	0.6177	0.5638	0.0772	0.0720
Spiral	MultiK	39.84	1.1184	1.0319	0.1398	0.1310
Spiral	MultiHash	3.96	0.0055	0.0052	0.0007	0.0007
Noisy annulus	ScalarK	5.29	0.0228	0.0220	0.0029	0.0027
Noisy annulus	MultiK	10.18	0.0619	0.0587	0.0077	0.0072
Noisy annulus	MultiHash	11.06	0.0478	0.0475	0.0060	0.0056
Rare disks	ScalarK	23.67	0.3659	0.3535	0.0457	0.0433
Rare disks	MultiK	33.50	0.6631	0.6261	0.0829	0.0784
Rare disks	MultiHash	38.22	0.6920	0.6527	0.0865	0.0799

Variance columns are raw logit variance, not probability MSE. The exact conditional mean of the 20-repeat ddof=0 variance estimator is 19/20 of population variance. Head L2 norms and per-record first-order sigmoid approximations are retained in the evidence; approximations do not replace the sampled probability diagnostics.

Joint measurement verification and costs

Each setting is sampled as one four-outcome multinomial. XX supplies correlated X0 and X1; ZZ supplies correlated Z0, Z1 and Z0Z1. Separate setting draws are independent. This preserves shared-shot covariance and applies the head to the resulting joint moments.

Coverage is 192 checkpoints, 6,144 checkpoint–input pairs, 12,800 joint-count arrays, 409,600 four-outcome count vectors and 245,760 scalar probability values. The counts represent 1,132,462,080 measurement outcomes generated by classical multinomial sampling. No individual physical shots were taken; Aer, IBM services and cloud jobs were not used.

Producer inner work took 7.630s wall and 6.000s CPU. The full invocation took 186.638s wall; whole-process sampled CPU was 181.453s, including startup, verification, derived work and writes. These clocks have different boundaries. The simulation is ideal: device noise, calibration drift, readout error, hardware queue time and physical latency are outside scope.

Independent replay reproduced all 12,800 count arrays exactly and checked all 245,760 derived probabilities within the registered 2e-12 tolerance. It separately reconstructed ideal joint probabilities (maximum difference 4.996e-16) and each checkpoint's ideal output (3.608e-15). Count replay uses the retained normalized joint probabilities after that separate check, avoiding artificial count changes from harmless floating-point differences.

Independent audit work took 7.200s wall and 6.984s CPU after its release gate; its complete invocation took 169.952s wall and 167.844s process CPU through exit. Inner and outer clocks are nested, not additive. Every displayed probability statistic and head/variance aggregate was checked against the committed replay and diagnostic evidence.

Sources: producer SHOTS_COMMIT and 192 diagnostic files; independent shot replay/commit and cost receipts. The maximum individual prediction-replay difference is not retained; 2e-12 is its checked tolerance, not a measured maximum.

Training costs and execution

All 608 official fits completed: 32 engineering, 64 development (A) and 512 confirmation (B). The training ledger records zero failed fits and zero retries. Each stage reached five active workers; all 608 worker receipts verify one effective BLAS thread and CPU-only execution.

Stage	Fits	Stage wall s	Worker CPU sum s	Trainer wall sum s
Engineering	32	12.26	48.36	31.34
A	64	1,501.24	7,200.36	7,152.40
B	512	16,643.28	82,002.94	79,355.59

Stage wall is the supervision interval. Worker CPU sums consumed process time through sampling. Trainer wall includes validation and loop evidence writes; simultaneous fit timers overlap.

Confirmation cost by model

Arm	Fits	Worker CPU sum s	Median trainer wall s
ScalarK	64	24,898.62	385.42
MultiK	64	26,624.59	416.23
MultiHash	64	26,700.61	404.06
TrigK	64	893.83	11.64
TrigHash	64	892.95	12.35
TrigG25	64	643.98	7.28
TrigG121	64	692.45	7.87
MLPG129	64	655.89	6.95

The largest individual worker peak was 61.76 MiB. This is not simultaneous system memory, and individual peaks must not be added. ETEL's logical groups of 64 (and peak ready capacity of 192) are bookkeeping, not active cores or a 64x/192x speedup. There is no matched serial-run estimate of parallel speedup.

Verification scoring and accounting limits

The following work is separate from the training totals. Both B audits verified 512 endpoints; official scoring and independent replay each covered all 512 selected models. Different timer boundaries are retained rather than combined into one misleading elapsed total.

Activity	Wall s	CPU s	Clock scope
Author B audit	105.93	158.16	Audit-body wall; whole-process CPU sample
Independent B audit	462.00	456.06	Child audit wall; leaf CPU through exit
Official scoring	32.88	28.45 / 187.89	Inner CPU / whole-process CPU sample
Independent score replay	80.85	79.86	Child replay wall; leaf CPU through exit

Coordinator launch-to-supervision gaps were 0.67s (engineering), 2.10s (A) and 16.61s (B), outside stage-wall entries. Worker CPU includes imports and pre-ownership checks; its ownership wall clock does not. Both stop before final resource/terminal serialization and process teardown.

Official scoring's inner timer starts after its release/hash gate: 28.45s inner CPU versus 187.89s whole-process sampled CPU. Their difference includes work outside the inner timer; it is not an independently measured gate-only cost. Independent audit/replay CPU receipts cover leaf processes through exit and exclude launcher, coordination wrapper and shell CPU. The author's inherited 'fit child' resource label refers here to its audit process.

Preserved operational failure

The first independent-score-replay wrapper invocation failed before launching its numerical child because the log parent directory was missing. The failure receipt was preserved. Recovery created that parent only and reused the unchanged command and still-unused output. This was an operational prelaunch failure, separate from the zero training failures. Its wrapper CPU cost was not recorded and is not silently treated as zero.

These tables exclude synthetic qualification, data preparation, inter-stage checks, reporting and later deployment phases. Allocated shared preparation charges are not added to measured CPU or wall totals. No energy, FLOP, GPU or physical-quantum speed claim follows. Worker metadata is checked against durable terminal events; trainer timers are bound by endpoint commits. Deployment costs are reported separately.

Sources: committed resource capsule and peer review; B author/independent audit receipts; committed scoring/replay summaries and leaf-process cost receipts; preserved prelaunch failure.

Earlier checks and remaining cost coverage

The earlier preparation and audit receipts provide additional measurements below. These clocks cover different scopes and are not added to training, deployment or one another. A missing CPU value is unmeasured, not zero.

Activity	Recorded wall s	Recorded CPU s
Official data preparation	71.019	Not recorded
Historical recheck within data preparation	50.363	Not recorded
Independent official data replay	20.534	Not recorded
Engineering author audit	0.677	1.250 whole-process sample
Engineering independent audit	1.516	Not recorded
Development A author audit	7.458	11.250 whole-process sample
Development A independent audit	30.472	Not recorded

The historical recheck is nested in data preparation. Author audit CPU samples include work before their internal wall clocks; these columns therefore have different boundaries. The retained cost scope note also lists 15 synthetic concurrency-qualification fits, 19 readiness or helper checks, rounded log-only observations and receipts without usable cost fields.

No complete measurement of manual effort, report authoring and rendering, all launcher and shell work, or operating-system overhead is available. Energy and thermal causes of timing drift were not measured. The package preserves available receipts without turning missing measurements into an all-in compute estimate.

Detailed sources and clock definitions: report_work/final_report/COST_SCOPE_NOTE_v01.json. Administrative formatting failures are retained in report preparation receipts and do not replace scientific attempts.

Architecture limits and next research decisions

The supported claim is specific: on these four synthetic task families, geometric grouping improved the registered local models over matched-count random routing. The experiment separates this effect from bank count more clearly than v0.6. It still changes which examples reach each bank, and jointly varies data and router fits across populations. It does not isolate every cause of specialization.

The stronger global Trig121 results show that basis bandwidth and function class remain central. A 121-parameter global model and a 125-parameter routed model have different features and optimization geometry. The comparison does not establish that routing is generally inferior or that a circuit can never help; it limits the claim for this fixed two-qubit, two-layer construction.

The complete architecture remains largely untested. Semantic priority ratings, adaptive routing and replay coupling, accepted and candidate histories, continual learning, sequence modeling, language-model scale and large-qubit behavior were not exercised. These small binary tasks cannot support comparison with general AI systems. Fresh draws from the same formulas do not establish external or out-of-distribution generalization.

Recommended focus for v08

Predeclare one distinctive architecture mechanism, such as adaptive bank allocation or priority-driven replay, and compare it with a frozen geometric route, matched random routing and a sufficiently flexible global classical control. Keep bank capacity, available information and training work explicit. Include a circuit-free local core to distinguish the mechanism from the two-qubit parameterization. Reserve fresh confirmation populations before tuning.

Recommended focus for v09 and the paper

Use a locked configuration on a broader task family or real benchmark, with controlled distribution shifts, imbalance and a quality-versus-cost curve. Select the candidate using development data only, then report all registered confirmation outcomes. Neither v0.8 nor v0.9 has been launched by this study.

For paper v1.0, present classical two-qubit emulation, the matched routing experiment, the exact classical representation change and the limits exposed by stronger controls. Treat original-architecture mechanisms as research directions until tested. For v1.1, mirror frozen circuits, heads and input routing on actual hardware with an explicit total-shot and latency budget; simulator parity alone does not predict physical-device performance.

Reproducibility and evidence

The delivered evidence package retains frozen protocols, fresh data and router policies, complete fit attempts, selected and final checkpoints, execution ledger evidence, official predictions, independent replays, derived representations, timing trials, simulated joint counts and report sources. Original absolute paths remain provenance; the delivery manifest inventories relative paths and file hashes.

Evidence	Relative location
Registered science	governance/STUDY_PLAN.json and ANALYSIS_PROTOCOL.json
Matrix and data	governance/RUN_MATRIX.json and quantum/data
Execution and complete fits	execution/state and execution/attempts
Official held-out results	analysis/official_results/stage_B_v01
Independent score replay	analysis/independent_results/stage_B_v01
Conversion and timing producers	analysis/deployment_results
Independent conversion and shots	analysis/independent_deployment_results
Independent timing summaries	analysis/independent_timing_results
Full report tables and source records	report_work/final_report
Portable inventory and metric verifier	reproducibility/verify_package.py

From the delivered package directory, verify its inventory and archived scores with a compatible Python interpreter:

```
python -B reproducibility/verify_package.py --package "C:\Q2\Improvement_Study_v0.7" --replay-scores
```

This read-only verifier checks inventory hashes and reconstructs metrics and the registered comparisons from archived predictions. It does not retrain the models or independently recreate predictions; separate retained numerical audits perform those prediction checks. The environment is documented rather than bundled, and portable fresh-root retraining has not been demonstrated.

Final acceptance is recorded after root and independent verification of the delivered inventory and report hash. The acceptance record is held outside the payload inventory to avoid circular self-hashes. The final report is released only after preservation checks and every-page Word rendering review pass.

Prior study preservation

The final preservation pass rehashed all 20,859 delivered v0.6 payloads, their root metadata and all four Baseline6 seal files. Every value matched the before-study record, including agreement between the archive, external manifest and companion hash. Prior study files were read only.

Preservation evidence: governance/PRIOR_PRESERVATION_BEFORE.json and governance/PRIOR_PRESERVATION_AFTER.json. Delivery acceptance remains separately recorded outside the package inventory after both destination verifiers pass.

Report edition and rights

Study v0.7. Report edition 1 prepared on 29 October 2026.

Anthony Siljeg — Founder, Alnotrea Labs · Company affiliation: Alnotrea Pty Ltd · Q-Pathformer Initiative. Anthony is the licensor; company affiliation is not an ownership transfer.

All original report paragraphs, tables, figures, numerical findings, limitations and historical recommendations are retained without alteration. Original internal labels identify the historical source edition. This appended note and document metadata are the only editorial changes. No scientific result was recalculated or removed.

Historical local paths, host configuration, execution and usage accounting identify retained evidence. The full archive remains separately preserved under C:\Q2. This curated release does not include raw datasets, trained checkpoints, installed environments or full operational archives. Retained archive sizes are not download sizes or hardware minimums.

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